

Qatar-1b

R-2802

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140723 · created 2026-09-05T00:02:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456861.7543 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.056
Transit depth (Rp/Rs)^2	3.71 +/- 2.17 [%]
Orbital Inclination (inc)	88.01 +/- 2.81
Airmass coefficient 1 (a1)	3.5 +/- 1.86
Airmass coefficient 2 (a2)	-0.85 +/- 1.58
Scatter in the residuals of the lightcurve fit is	59.31 %
Best Comparison Star	#3 - [56, 107]
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.0819 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.056 vs 0.1463 (deviation 0.83σ)
Duration fit vs published	0.0819 d vs 0.0692 d (deviation 1.45σ)
Mid-time O–C	0.5 min
Depth SNR	3.4
Residual scatter	59.31 %

Light curve

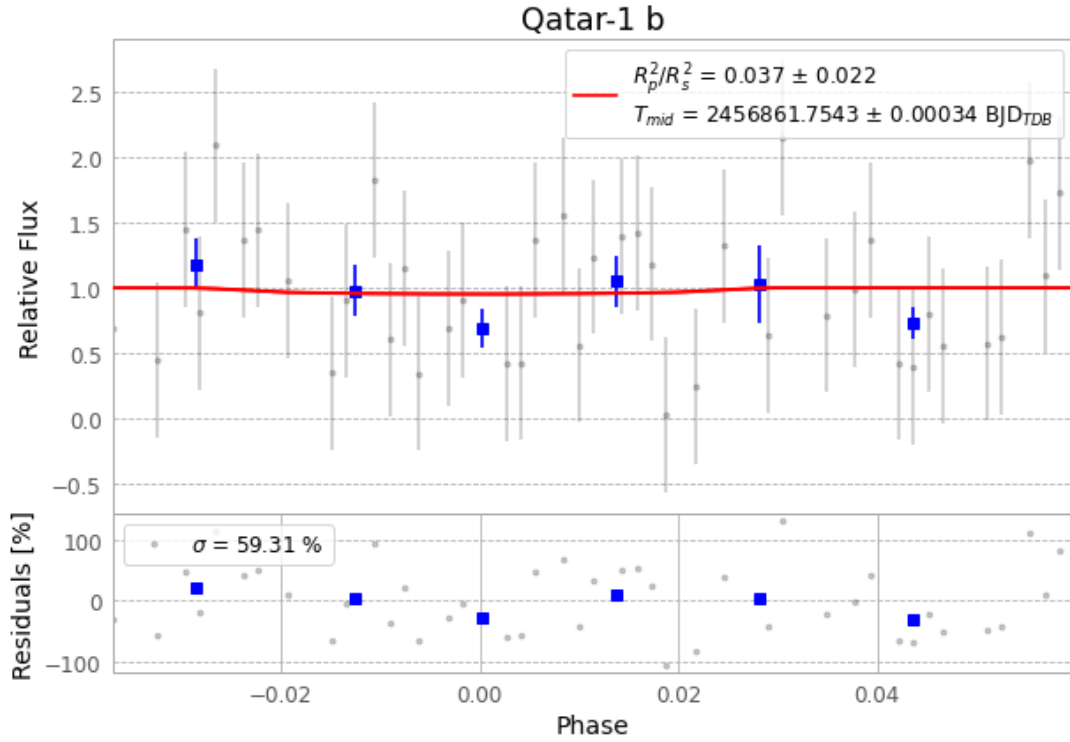


Figure 1 — FinalLightCurve_Qatar-1 b_2014-07-22.png (SHA-256 479de7b2c82c05bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d19b6262e3cfdd6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), Qatar-1140723044513.FITS → Qatar-1140723080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140723.log (d88af718d695...), FinalLightCurve_Qatar-1 b_2014-07-22.png (479de7b2c82c...), FinalLightCurve_Qatar-1 b_2014-07-22.csv (d02556d986ac...)

Compute resources

Wall time 334.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:02Z.